Keyan Vakil

Senior Software & Data Engineer

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· https://github.com/KeyanVakil

Professional Profile

Software and data engineer with over six years of experience in backend development, including API design and implementation using C# and .NET. Skilled in SQL for database design, data modeling, and performance optimization on large datasets, with a focus on secure and efficient solutions in regulated environments. Experienced in CI/CD pipelines, automation scripting in Python and PowerShell, and collaborating in distributed teams to deliver reliable integrations and migrations.

Demonstration Project

Agentic AI Code Review Platform Apr 2026

Built for Capgemini Norge AS — AI-Utvikler Role

https://github.com/KeyanVakil/capgemini-norge-as$_ai - utvikler_4$58002275

- Built a multi-agent code review platform in Go that orchestrates four specialized AI agents (security, style, test generation, improvement synthesis) using goroutines and channels for parallel execution with DAG-based workflow coordination.
- Integrated the Claude API (Anthropic) with streaming SSE responses, structured output extraction, and tailored system prompts per agent — demonstrating production-grade agentic AI pipelines aligned with Capgemini's AI-integrated development focus.
- Containerized the full stack with Docker Compose (Go backend, PostgreSQL 16 with JSONB storage, automated migrations) for single-command deployment, with GitHub Actions CI running lint, unit tests, and integration tests against a real database.
- Designed a real-time web UI with Server-Sent Events that visualizes the agent workflow as it executes, streaming each agent's progress and findings to the browser as they complete.

Work Experience

International Travel & Professional Development Sabbatical Oct 2025 – Feb 2026

Worldwide

- Intentional break for global travel and personal renewal.
- Experimented hands-on with generative coding agents (Claude Code, OpenAI APIs) and Python automation to explore agentic AI pipelines, agentic RAG for enhanced retrieval/reasoning, context engineering, and Model Context Protocol (MCP) integrations for tool orchestration and multi-agent interoperability.
- Focused on 2025–2026 trends: autonomous multi-agent systems, production-scale AI-assisted engineering, and multimodal/embodied AI foundations to stay aligned with the shift toward agentic workflows and intent-driven development.

Senior Conversion Engineer (Software/Data) Aug 2022 – Aug 2025

Tyler Technologies — Remote (USA)

Large-scale data migrations and internal tooling development for property assessment and tax systems across multiple jurisdictions.

- Developed SQL-based data conversions to import and transform data from legacy property tax valuation and assessment systems, using query rewriting, cursor tuning, indexing, and performance analysis to achieve runtime improvements (often 5–10x faster on enterprise-scale datasets).

- Built PowerShell and Python automation for SQL Server to Oracle migrations, reducing process time from 3–5 days to around 24 hours on datasets up to 200k parcels and over 100 GB.
- Implemented pure T-SQL solutions for complex geospatial conversions (500k coordinates from GIS dumps), including tolerance alignment, direction logic, and validation for high automated ingest rates.
- Automated database cleanup processes for new clients using Python and OpenAI API, reducing manual effort from 4 hours to under 1 hour.

Software Engineer (Backend) Mar 2020 – Aug 2022

California State Lottery — Sacramento, CA

Backend development in a government-regulated financial environment handling sensitive personal and financial data across finance, HR, and operations.

- Received California State Lottery "Superior Accomplishment Award" in 2021.
- Participated in the upgrade from Epicor 7.4 to 10, collaborating with accountants, auditors, database administrators, project managers, and contractors.
- Co-maintained Data Integration Manager (C#/.NET ETL tool) for integrations with banking partners and internal systems; also developed internal web applications using JavaScript, HTML/CSS, C#/.NET Framework, Entity Framework, and MySQL.
- Implemented Azure DevOps CI/CD pipelines, contributing to improved release traceability and fewer configuration issues.
- Developed prototype for PII/SSN obfuscation using cryptographic salting and object-oriented design to enable compliant test datasets.

Software Engineer (Full-Stack) Jul 2019 – Mar 2020

Outploy Inc. — Folsom, CA

Healthcare SaaS (dental practice management) on a PHP/MySQL monolith under HIPAA-regulated constraints.

- Contributed to the gradual transition of the web application from a custom jQuery framework to Vue.js, applying object-oriented concepts in component-based development while updating legacy workflows.
- Participated in shifting from monolithic to microservices architecture, developing new Python-based REST endpoints.
- Implemented secure asynchronous document upload with short-lived tokens, direct-to-S3 storage, and IAM controls.
- Applied security measures compliant with HIPAA, including TLS, encryption at rest, RBAC, and audit logging.

Technical Skills

Languages: Python, C#, Go, SQL (T-SQL, PL/SQL), PowerShell, JavaScript, PHP, HTML/CSS
Databases: SQL Server, Oracle, MySQL, PostgreSQL
Cloud & DevOps: AWS (S3), Azure, Azure DevOps, Docker, CI/CD Pipelines, GitHub Actions
Frameworks & Tools: .NET, Vue.js, REST APIs, ETL Development, Git
AI & Automation: Claude API, OpenAI API, Agentic AI Pipelines, MCP Integrations
Domains: Data Migrations, Automation, Compliance (HIPAA, PII Handling)

Education

Bachelor's Degree, Computer Science Sept 2014 – May 2018

Boston University — Boston, MA

- Dean's List · GPA: 3.5/4.0

Languages

English: Native · **Spanish:** Conversational · **Norwegian:** Beginner, Actively Studying